

ABES ENGINEERING COLLEGE, GHAZIABAD

Department of Applied Sciences and Humanities

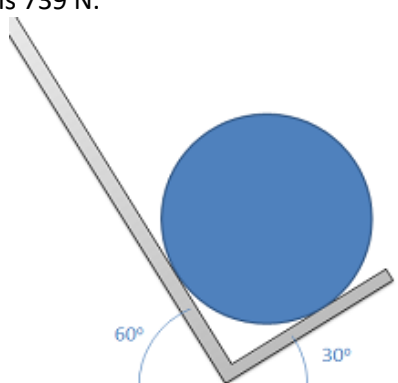
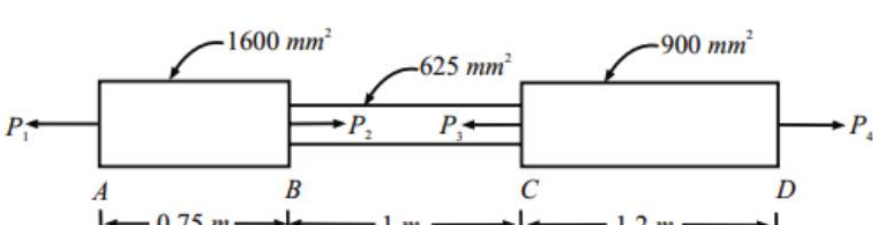
Basics of Mechanical and Automation Engineering (25ME201)

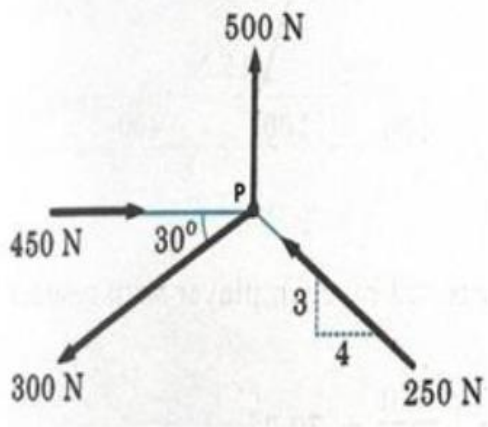
Assignment-1 Unit-I (Introduction to Mechanics and Strength of Materials)

Date of sharing: 26/02/2026

Last Date of Submission: 12/03/2026

Section: All

S. No.	CO	Questions	K Level
1	CO1	A roller of weight 233 kN rests in an L-shaped support-frame as shown in the figure. Determine the contact reactions at all the points of contact. Weight of the roller is 739 N. 	K3
2	CO1	A simply supported beam AB of span 10 m is hinged at A and roller supported at B. The beam carries: • A uniformly varying load (UVL) increasing linearly from 0 kN/m at A to 8 kN/m at B • A point load of 20 kN acting 3 m from A Draw the Free Body Diagram and determine the reactions at supports A and B.	K3
3	CO1	A member is subjected to point loads P_1, P_2, P_3 and P_4 as shown in figure. Calculate the force P_3 necessary for equilibrium if $P_1=110$ kN, $P_2=210$ kN and $P_4=150$ kN. Determine change in length of the member. Take moduli of elasticity 210 GN/m². 	K3

4	CO1	A system of four forces acting on a body is as shown in figure. Determine the magnitude resultant and its inclination with positive X-axis. 	K2
5	CO1	With the help of a neat diagram, draw the stress–strain curve for mild steel and explain the salient features of each point on the curve.	K2